

17. Together with W. Schmeidler: Modules in Noncommutative Domains, Especially from Differential and Difference Expressions

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§ 10. Existence of Infinitely Many Decompositions of a Group.

If the representation of a completely reducible group as a sum is not unique, then by Theorem V every representation contains at least two mutually isomorphic subgroups. We now consider the subgroup formed from such a system of isomorphic subgroups,

$$\mathfrak{H} = \mathfrak{A}_1 + \cdots + \mathfrak{A}_\alpha,$$

where $\mathfrak{A}_1, \mathfrak{A}_2, \dots, \mathfrak{A}_\alpha$ are all isomorphic, and assert that $\mathfrak{H}$ then admits infinitely many distinct decompositions of this kind, provided only that the rationality domain P contains infinitely many “constants”, that is, quantities a for which $\xi_i a = a \xi_i$. This theorem is valid even when no further hypothesis is imposed on the groups $\mathfrak{A}_i$, not even irreducibility. Thus we have:

Theorem VIII. *If $\mathfrak{G}$ can be represented as a sum of finitely many mutually isomorphic subgroups, then this representation is possible in infinitely many distinct ways, provided only that the rationality domain P contains infinitely many constants. Here constants mean those quantities a of the domain for which $\xi_i a = a \xi_i$.*

Proof. Let

$$\mathfrak{G} = \mathfrak{A}_1 + \cdots + \mathfrak{A}_\alpha,$$

and let $\mathfrak{A}_i$ be isomorphic to $\mathfrak{A}_\kappa$ for $i, \kappa = 1, \dots, \alpha$ ($\alpha \geq 2$). In order to specify a definite isomorphic assignment, we first single out one subgroup, say $\mathfrak{A}_1$. Let the isomorphism between $\mathfrak{A}_1$ and $\mathfrak{A}_i$ be fixed by

$$a_1 \sim t_{1i} a_i, \quad a_i \sim t_{i1} a_1, \quad t_{11} a_1 = a_1 \quad (i, \kappa = 1, \dots, \alpha), \quad (31)$$

so that, in the group $\mathfrak{A}_1$, each residue class is assigned to itself. Since $Ma_1 = 0$ and $Ma_i = 0$, it follows that $Mt_{1i} a_i = 0$ and $Mt_{i1} a_1 = 0$; hence the classes $t_{1i} a_i$ and $t_{i1} a_1$, analogously to units, admit multiplication by classes, not merely by polynomials. It follows from the definition of isomorphism that, for two assigned classes $\mathfrak{h}$ and $\bar{\mathfrak{h}}$ admitting this class multiplication, the classes $\mathfrak{r}\mathfrak{h}$ and $\mathfrak{r}\bar{\mathfrak{h}}$ are again assigned to each other. Therefore the relations $a_r a_s = 0$ ($r \neq s$) ($r = 1, \dots, \alpha$; $s = 1, \dots, \alpha$) give, for $s = 1$,

$$\left\{ \begin{array}{ll} a_r t_{1\kappa} a_\kappa = 0 & (\kappa = 1, \dots, \alpha, r \neq 1), \\ \text{and for } s > 1 & a_r t_{s1} a_1 = 0 \quad (r \neq s), \\ \text{hence also} & t_{1i} a_i = a_1 t_{1i} a_i, \quad t_{i1} a_1 = a_i t_{i1} a_1. \end{array} \right. \quad (32)$$

Furthermore, from (31), generalizing (27) and (28), the uniqueness of the assignment gives

$$t_{1i} t_{i1} a_1 = a_1, \quad t_{\kappa 1} t_{1\kappa} a_\kappa = a_\kappa. \quad (33)$$

Let the isomorphism between $\mathfrak{A}_i$ and $\mathfrak{A}_\kappa$ now be the relation obtained by composing (31); such a stipulation is necessary, since the individual groups may admit isomorphisms into themselves. From (31), (32), and (33) we therefore get

$$a_i \sim t_{i1} t_{1\kappa} a_\kappa = t_{i\kappa} a_\kappa, \quad t_{i\kappa} a_i = a_i, \quad a_r t_{s\kappa} a_\kappa = 0 \quad (r \neq s), \quad (34)$$

where $t_{i1} t_{1\kappa} = t_{i\kappa}$ has been put. From (33) and (34) one obtains further

$$t_{i\kappa} t_{\kappa l} a_l = t_{i1} t_{1\kappa} t_{\kappa l} a_l = t_{i1} t_{1l} a_l = t_{il} a_l;$$

thus the singling out of the group $\mathfrak{A}_1$ is again removed. The isomorphic relation between $\mathfrak{A}_i$ and $\mathfrak{A}_\kappa$ remains the same, whichever group $\mathfrak{A}_\kappa$ is used in place of $\mathfrak{A}_1$ for the definition. Hence the $t_{i\kappa}$ are classes which, in summary, satisfy the relations

$$Mt_{i\kappa} a_\kappa = 0; \quad a_r t_{i\kappa} a_\kappa = 0 \quad (r \neq i); \quad t_{i\kappa} t_{\kappa l} a_l = t_{il} a_l; \quad t_{\kappa i} a_i = t_{\kappa l} t_{li} a_l = a_\kappa. \quad (35)$$

These relations say precisely that the modules $\mathfrak{M}_i$ and $\mathfrak{M}_\kappa$ belonging to $\mathfrak{A}_i$ and $\mathfrak{A}_\kappa$ are of the same kind (cf. the definition in § 9); in place of the P and Q there we have here the classes $t_{i\kappa}$ and $t_{\kappa i}$, and the first

two relations (35), taken together, correspond to formula (25) or (26). Conversely, these relations imply the isomorphism by Theorem VI.²⁰

From the classes $t_{i\mathfrak{x}}a_{\mathfrak{x}}$ we now compose classes b_ρ which will again turn out to be units of groups $\mathfrak{B}_\rho$, whence a representation

$$\mathfrak{G} = \mathfrak{B}_1 + \cdots + \mathfrak{B}_\alpha$$

will follow. Let $\lambda_{i\mathfrak{x}}$ be constants whose determinant $|\lambda_{i\mathfrak{x}}|$ is nonzero, and let $\lambda^{\mathfrak{x}r}$ be the corresponding cofactors divided by $|\lambda_{i\mathfrak{x}}|$, so that the relations

$$\begin{cases} \sum_{r=1}^{\alpha} \lambda_{ri} \lambda^{\mathfrak{x}r} = \delta_{i\mathfrak{x}} = \begin{cases} 0, & i \neq \mathfrak{x}, \\ 1, & i = \mathfrak{x}, \end{cases} \\ \sum_{r=1}^{\alpha} \lambda_{ir} \lambda^{\mathfrak{x}r} = \delta_{i\mathfrak{x}} \end{cases} \quad (36)$$

hold. We then define

$$b_\rho = \sum_{i,\mathfrak{x}} \lambda_{\rho i} \lambda^{\rho\mathfrak{x}} t_{i\mathfrak{x}} a_{\mathfrak{x}} \quad (\rho = 1, \dots, \alpha). \quad (37)$$

From (35), (36), and (37) one obtains $Mb_\rho = 0$ and

$$\sum_{\rho=1}^{\alpha} b_\rho = \sum_{i,\mathfrak{x},\rho} \lambda_{\rho i} \lambda^{\rho\mathfrak{x}} t_{i\mathfrak{x}} a_{\mathfrak{x}} = \sum_{i,\mathfrak{x}} \delta_{i\mathfrak{x}} t_{i\mathfrak{x}} a_{\mathfrak{x}} = \sum_i t_{ii} a_i = \sum_i a_i = e, \quad (38)$$

and also

$$\begin{aligned} b_\rho b_\sigma &= \sum_{\substack{i,\mathfrak{x} \\ \mu,\nu}} \lambda_{\rho i} \lambda^{\rho\mathfrak{x}} \lambda_{\sigma \mu} \lambda^{\sigma\nu} t_{i\mathfrak{x}} a_{\mathfrak{x}} t_{\mu\nu} a_\nu \\ &= \sum_{i,\mu,\nu} \lambda_{\rho i} \lambda^{\rho\mu} \lambda_{\sigma \mu} \lambda^{\sigma\nu} t_{i\nu} a_\nu = \delta_{\rho\sigma} \sum_{i,\nu} \lambda_{\rho i} \lambda^{\sigma\nu} t_{i\nu} a_\nu = \begin{cases} 0 & (\rho \neq \sigma), \\ b_\rho & (\rho = \sigma), \end{cases} \end{aligned}$$

$(\rho, \sigma = 1, \dots, \alpha)$. Thus the representation $\mathfrak{r}e = \mathfrak{r}b_1 + \cdots + \mathfrak{r}b_\alpha$ is an additive decomposition of the group $\mathfrak{G}$. The groups $\mathfrak{B}_\rho$ are distinct from the groups $\mathfrak{A}_\sigma$ as soon as the $\lambda_{i\mathfrak{x}}$ do not merely form a diagonal matrix. For if we form $a_\sigma b_\rho$, then by (35) we get

$$a_\sigma b_\rho = \sum_{i,\mathfrak{x}} \lambda_{\rho i} \lambda^{\rho\mathfrak{x}} a_\sigma t_{i\mathfrak{x}} a_{\mathfrak{x}} = \lambda_{\rho\sigma} \sum_{\mathfrak{x}} \lambda^{\rho\mathfrak{x}} t_{\sigma\mathfrak{x}} a_{\mathfrak{x}} \neq 0 \quad \text{for } \lambda_{\rho\sigma} \neq 0,$$

since otherwise each individual summand in the last sum would have to vanish, which is not the case. Thus if, for fixed σ , there are at least two ρ with $\lambda_{\rho\sigma} \neq 0$, then $\mathfrak{A}_\sigma$ cannot be identical with any $\mathfrak{B}_\rho$.

But the groups $\mathfrak{B}_\rho$ are also mutually isomorphic and isomorphic to the groups $\mathfrak{A}_\sigma$. We first show the existence of classes $r_{\rho\sigma}a_\sigma$ and $r^{\sigma\rho}b_\rho$ which mediate the isomorphism of $\mathfrak{A}_\sigma$ and $\mathfrak{B}_\rho$ ($\rho, \sigma = 1, \dots, \alpha$). Indeed, we show that for the classes

$$r_{\rho\sigma}a_\sigma = \sum_i \lambda_{\rho i} t_{i\sigma} a_\sigma, \quad r^{\sigma\rho} = \sum_{\mathfrak{x}} \lambda^{\rho\mathfrak{x}} t_{\sigma\mathfrak{x}} a_{\mathfrak{x}} = r^{\sigma\rho}b_\rho$$

the relations “of the same kind” are fulfilled between the modules

$$(\mathfrak{M}, b_1 + \cdots + b_{\rho-1} + b_{\rho+1} + \cdots + b_\alpha)$$

and

$$\begin{aligned} &(\mathfrak{M}, a_1 + \cdots + a_{\sigma-1} + a_{\sigma+1} + \cdots + a_\alpha) : \\ &\begin{cases} b_{\mathfrak{x}} r_{\rho\sigma} a_\sigma = 0 & (\mathfrak{x} \neq \rho), & r^{\sigma\rho} r_{\rho\sigma} a_\sigma = a_\sigma, \\ a_{\mathfrak{x}} r^{\sigma\rho} b_\rho = 0 & (\mathfrak{x} \neq \sigma), & r_{\rho\sigma} r^{\sigma\rho} b_\rho = b_\rho, \end{cases} \end{aligned} \quad (39)$$

from which, as in (35), the isomorphism follows through the assignment

$$b_\rho \sim r_{\rho\sigma} a_\sigma, \quad a_\sigma \sim r^{\sigma\rho} b_\rho. \quad (40)$$

²⁰And in fact without applying the finiteness theorem, since the formulas already display respectively a map $\varphi_{i\mathfrak{x}}$ and its inverse $\varphi_{\mathfrak{x}i}$. For from $ra_i = ra_i t_{i\mathfrak{x}} a_{\mathfrak{x}} t_{\mathfrak{x}i} a_i$ it follows that $ra_i \neq 0$ also implies $ra_i t_{i\mathfrak{x}} a_{\mathfrak{x}} \neq 0$.

Indeed,

$$\begin{aligned} b_{\varkappa} r_{\rho\sigma} a_{\sigma} &= \sum_{\mu, \nu, i} \lambda_{\varkappa\mu} \lambda^{\nu\mu} t_{\mu\nu} a_{\nu} \lambda_{\rho i} t_{i\sigma} a_{\sigma} \\ &= \sum_{\mu, i} \lambda_{\varkappa\mu} \lambda^{\mu i} \lambda_{\rho i} t_{i\sigma} a_{\sigma} = \delta_{\varkappa\rho} \sum_{\mu} \lambda_{\varkappa\mu} t_{\mu\sigma} a_{\sigma} = \delta_{\varkappa\rho} r_{\rho\sigma} a_{\sigma} = 0 \quad (\varkappa \neq \rho), \end{aligned}$$

and further

$$r^{\sigma\rho} r_{\rho\sigma} a_{\sigma} = \sum_{\nu, i} \lambda^{\rho\nu} t_{\sigma\nu} a_{\nu} \lambda_{\rho i} t_{i\sigma} a_{\sigma} = \sum_i \lambda^{\rho i} \lambda_{\rho i} a_{\sigma} = a_{\sigma}.$$

Conversely, in the same way one finds

$$a_{\varkappa} r^{\sigma\rho} b_{\rho} = a_{\varkappa} \sum_{\mu} \lambda^{\rho\mu} t_{\sigma\mu} a_{\mu} = 0 \quad (\varkappa \neq \sigma),$$

and

$$r_{\rho\sigma} r^{\sigma\rho} b_{\rho} = \sum_{i, \mu} \lambda_{\rho i} t_{i\sigma} a_{\sigma} \lambda^{\rho\mu} t_{\sigma\mu} a_{\mu} = \sum_{i, \mu} \lambda_{\rho i} \lambda^{\rho\mu} t_{i\mu} a_{\mu} = b_{\rho}.$$

From the relations of the same kind just proved, the isomorphism of $\mathfrak{A}_{\sigma}$ and $\mathfrak{B}_{\rho}$ follows (again without the finiteness theorem). The isomorphism between $\mathfrak{B}_{\rho}$ and $\mathfrak{B}_{\tau}$ is obtained from this by composition; we record the formulas. Put

$$\ell_{\rho\tau} b_{\tau} = r_{\rho\sigma} r^{\sigma\tau} b_{\tau} = \sum_{i, j} \lambda_{\rho i} t_{i\sigma} a_{\sigma} \lambda^{\tau j} t_{\sigma j} a_j = \sum_{i, j} \lambda_{\rho i} \lambda^{\tau j} t_{ij} a_j;$$

then the assignment is $b_{\rho} \sim \ell_{\rho\tau} b_{\tau}$. Here too the relations of the same kind can again be verified formally; if the $\ell_{\rho\tau}$ are put in place of the $t_{i\kappa}$, they satisfy (35). Indeed,

$$b_{\sigma} \ell_{\rho\tau} b_{\tau} = b_{\sigma} r_{\rho\sigma} r^{\sigma\tau} b_{\tau} = 0 \quad (\sigma \neq \rho),$$

$$\ell_{\rho\sigma} \ell_{\sigma\tau} b_{\tau} = r_{\rho\kappa} r^{\kappa\sigma} r_{\sigma\lambda} r^{\lambda\tau} b_{\tau} = r_{\rho\kappa} a_{\kappa} r^{\kappa\tau} b_{\tau} = \ell_{\rho\tau} b_{\tau}.$$

This proves Theorem VIII, since one need only choose the $\lambda_{i\kappa}$ in infinitely many ways so that they form no diagonal matrix and also are not transformed into one another by diagonal matrices.

A consequence of Theorem VIII, in connection with Theorem V or VII, is:

Theorem IX.²¹ *Let the module $\mathfrak{M}$ be completely reducible and the least common multiple of prime modules $\mathfrak{P}_1, \dots, \mathfrak{P}_k$ which form a totally relatively prime system, and suppose the rationality domain P contains infinitely many constants. Then the necessary and sufficient condition that $\mathfrak{M}$ be representable in infinitely many distinct ways as the least common multiple of prime modules is that at least two of the modules $\mathfrak{P}_1, \dots, \mathfrak{P}_k$ be of the same kind.*

We shall now derive general criteria for a module $\mathfrak{M}$ to be divisible by infinitely many prime modules, and first prove the following:

Auxiliary theorem.²² *If a completely reducible module $\mathfrak{M} = [\mathfrak{P}_1, \dots, \mathfrak{P}_k]$ is divisible by a prime module $\mathfrak{P}'$ different from all the $\mathfrak{P}_i$, then at least two of the $\mathfrak{P}_i$ are of the same kind.*

We first select from $\mathfrak{P}_1, \dots, \mathfrak{P}_k$ a totally relatively prime system by successively deleting every one of these prime modules that is contained in the least common multiple of those not yet deleted. Hence assume now that $\mathfrak{P}_1, \dots, \mathfrak{P}_k$ themselves form a totally relatively prime system. Let a' be a class of $\mathfrak{M}$ corresponding to the unit class modulo $\mathfrak{P}'$. Then

$$Fa' = Fa'a_1 + \dots + Fa'a_k$$

for every polynomial F . At least two of the products $a'a_i$ are nonzero, say $a'a_1$ and $a'a_2$; for if, for instance, only $a'a_1 \neq 0$, then $Fa' = Fa'a_1$, and the residue classes Fa' would form a subgroup of $\mathfrak{A}_1$, hence, since $\mathfrak{A}_1$ is a prime group, would be identical with $\mathfrak{A}_1$, so that the modules $\mathfrak{P}'$ and $\mathfrak{P}_1$ would also coincide. By the reasoning of § 8, it follows that $\mathfrak{A}'$ is isomorphic to $\mathfrak{A}_1$ and $\mathfrak{A}_2$.

Let $\mathfrak{M}$ now be an arbitrary module. We consider the least common multiple $\mathfrak{N}$ of all prime modules by which $\mathfrak{M}$ is divisible.²³ Two cases are possible. Either $\mathfrak{N}$ is not representable as the least common multiple of finitely many prime modules; in this case $\mathfrak{N}$, and hence also $\mathfrak{M}$, has infinitely many prime modules as divisors. Or $\mathfrak{N}$ is representable as the least common multiple of finitely many prime modules; then $\mathfrak{N}$ is called the

²¹ Cf. Loewy, pp. 106–107.

²² Cf. Loewy, p. 96.

²³ If there are infinitely many of these, then by Theorem IV no additive decomposition of the residue group can correspond to this representation. Cf. also the example § 12, 4.

greatest completely reducible module of $\mathfrak{M}$ (following a corresponding construction of Loewy). If among these finitely many prime modules two are of the same kind, then by Theorem IX the module $\mathfrak{N}$, and therefore $\mathfrak{M}$, is divisible by infinitely many prime modules. Conversely, if the latter is the case, then there is at least one prime divisor of $\mathfrak{N}$ different from all the finitely many modules whose least common multiple represents $\mathfrak{N}$. By the auxiliary theorem, $\mathfrak{N}$, hence $\mathfrak{M}$, has two prime divisors of the same kind. Thus we have proved:

Theorem X. *If the rationality domain P contains infinitely many constants, then the necessary and sufficient condition that a module be divisible by infinitely many distinct prime modules is that either no greatest completely reducible module exists, or, if such a module exists, that it be divisible by at least two distinct prime modules of the same kind.*

§ 11. Relations to Systems of Differential Equations and Their Integrals.

By virtue of the finiteness theorem, to every module of differential expressions there can be assigned as a basis a system of finitely many differential expressions, so that the totality of simultaneous integrals of all differential equations obtained by setting a differential expression of the module equal to zero coincides with the totality of integrals of the finite system of differential equations obtained by setting the basis differential expressions equal to zero. This gives the connection with the theory of systems of linear partial differential equations; we pursue this connection a little further in the case where the residue group of the module is “finite”.

Definition. *A residue group is called finite if there is a number μ , the order of the group, such that between any $\mu + 1$ residue classes there is a linear relation with coefficients from P , while there is at least one system of μ residue classes between which no relation exists, and which we shall call a linear basis of the residue group. In the other case the residue group is called infinite.*

The sum of two finite residue groups has finite order, namely the sum of their orders. The sum of two infinite groups, or of a finite and an infinite group, is an infinite group. Examples of finite groups are furnished by all residue groups of modules in one variable; but for several variables the order can also be finite, for instance for the module with basis ξ_1^2, ξ_2^3 , where all residue classes can be linearly composed from those of $1, \xi_1, \xi_2, \xi_1\xi_2$. For examples of infinite groups see § 12.

We now have the following:

Theorem XI. *If a module has a finite residue group, then every system of differential equations obtained by setting a basis of the module equal to zero admits exactly as many linearly independent integrals as the order of its residue group indicates. Conversely, for a given system of finitely many linear partial differential equations that possess only a finite number of linearly independent integrals, this number is equal to the order of the residue group of the module generated by the differential expressions.*

Here the rationality domain P consists of analytic functions of n variables $x_1, \dots, x_n$, with the occurring singularities, inside a certain n -dimensional domain, forming at most manifolds of lower dimensions, so that there are arbitrarily many points with regular n -dimensional neighborhoods.

Let $\mathfrak{M}$ be a module with finite residue group. We show that there is a linear basis of power products with the property that *when an arbitrary residue class is expressed through this basis, only powers of finitely many different quantities from P can occur in the denominator.*

For this purpose we order all power products $\xi_1^{a_1} \dots \xi_n^{a_n}$ lexicographically and take as representatives $Q_1, \dots, Q_m$ of the residue classes all those that are linearly independent modulo $\mathfrak{M}$ of the preceding ones. By hypothesis there are only finitely many such products. All the others can be expressed linearly through these.

Let $\mathfrak{S}$ be the system of all power products not admitted into the linear basis. Then, by the first part of the proof of Theorem III, $\mathfrak{S}$ has a finite subsystem $\mathfrak{T}$ of power products such that every power product from $\mathfrak{S}$ is divisible by at least one from $\mathfrak{T}$. Let $P_1, \dots, P_\rho$ be the power products of $\mathfrak{T}$, and let $M_1 = 0, \dots, M_\rho = 0$ ($\mathfrak{M}$) be the relations by means of which $P_1, \dots, P_\rho$ are each expressed modulo $\mathfrak{M}$ by a linear combination of lower power products; for example,

$$M_i = b_i P_i + \text{lower terms.}$$

If P is an arbitrary power product from $\mathfrak{S}$, hence of the form

$$\xi_1^{h_1} \dots \xi_n^{h_n} P_i,$$

then

$$\begin{aligned} \xi_1^{h_1} \dots \xi_n^{h_n} M_i &= b_i \xi_1^{h_1} \dots \xi_n^{h_n} P_i + \text{lower terms} \\ &= b_i P + \text{lower terms.} \end{aligned}$$

Assume now that it has already been proved for all power products lower than P that only powers of $b_1, \dots, b_\rho$ occur in denominators, while the numerators are formed from the coefficients of the M_i by addition, multiplication, and differentiation. Then the same holds for P itself. This assumption is permitted, since it is fulfilled for P_1 , the first element from $\mathfrak{S}$ and $\mathfrak{T}$.

By Theorem III, moreover, $M_1, \dots, M_\rho$ is a module basis of $\mathfrak{M}$. Now consider a system of values $\beta_1, \dots, \beta_n$ of $x_1, \dots, x_n$ for which $b_1 \neq 0, \dots, b_\rho \neq 0$ and the remaining coefficients of the M_i are regular, that is, a regular point of the system of differential equations $M_1 = 0, \dots, M_\rho = 0$. The power products $Q_i = \xi_1^{h_{i1}} \dots \xi_n^{h_{in}}$ ($i = 1, \dots, m$) correspond to the partial derivatives

$$\frac{\partial^{h_{i1} + \dots + h_{in}}}{\partial x_1^{h_{i1}} \dots \partial x_n^{h_{in}}} y,$$

for which we prescribe arbitrary constant values $(\gamma_1, \dots, \gamma_m)$ at the point $(\beta_1, \dots, \beta_n)$. Then the differential equations $M_i = 0$ determine uniquely, as finite values at this point, the values of all higher derivatives of y , since only the b_i occur in denominators; and it is known that this implies the existence of m linearly independent analytic integrals.

On the other hand, under our hypotheses there cannot exist more than m linearly independent integrals, since otherwise at a regular point one could prescribe arbitrarily the values of $m+1$ suitably chosen derivatives, contrary to the fact that linear dependencies modulo $\mathfrak{M}$ must exist among any $m+1$ power products.

Conversely, if a module has an infinite residue group, the same method shows that there are infinitely many linearly independent integrals; therefore, in the case of only m linearly independent integrals, the residue group is finite, and its order is, by the above, equal to m .

To close this paragraph we show that, for two modules of differential expressions, the concept of isomorphism, or equivalently of being of the same kind, has for the integrals of the associated systems of differential equations the same meaning as Poincaré's concept of kind for one variable (cf. Blumberg, loc. cit., Appendix I).

Theorem XII. *If two modules $\mathfrak{M}$ and $\mathfrak{N}$ of differential expressions are of the same kind, then there are two differential expressions P and Q such that $P(y)$ and $Q(z)$ run through all integrals of $\mathfrak{N}$ and $\mathfrak{M}$, respectively, when y runs through all integrals of $\mathfrak{M}$ and z through all integrals of $\mathfrak{N}$.*

By hypothesis, namely (by § 9), for two suitably chosen polynomials P and Q the relations

$$\begin{aligned} MQ &= 0(\mathfrak{N}), & PQ &= 1(\mathfrak{N}), \\ NP &= 0(\mathfrak{M}), & QP &= 1(\mathfrak{M}) \end{aligned}$$

hold. Thus, since $NP(y) = 0$, the function $P(y)$ is an integral of $\mathfrak{N}$. Likewise $Q(z)$ is an integral of $\mathfrak{M}$. Since $PQ(z) = z$ and $QP(y) = y$, $P(y)$ runs through all integrals of $\mathfrak{N}$, and $Q(z)$ all integrals of $\mathfrak{M}$. In this way, to each nonzero y there corresponds a nonzero z , and conversely.

§ 12. Examples.

1. As a first example we consider an ordinary linear differential equation of second order whose coefficients are represented as symmetric functions of the integrals. The rationality domain P consists here of all rational functions of the integrals and their derivatives; we restrict ourselves to a sufficiently small neighborhood of a regular point of the differential equation. We define the module $\mathfrak{M}$ as the totality of all differential expressions that have y_1 and y_2 as integrals, and for which we again write polynomials in one indeterminate ξ . The module basis is then one-membered, and is given by the polynomial

$$M = \begin{vmatrix} y_1 & y_2 \\ y'_1 & y'_2 \end{vmatrix} \xi^2 - \begin{vmatrix} y_1 & y_2 \\ y''_1 & y''_2 \end{vmatrix} \xi + \begin{vmatrix} y'_1 & y'_2 \\ y''_1 & y''_2 \end{vmatrix},$$

which is uniquely determined up to a factor from P . Now

$$\mathfrak{M} = [\mathfrak{M}_1, \mathfrak{M}_2], \tag{41}$$

where $\mathfrak{M}_i$ consists of all differential expressions having the integral y_i , and whose basis is determined up to a factor from P by

$$M_i = y_i \xi - y'_i \quad (i = 1, 2).$$

Indeed, $\mathfrak{M}$ is divisible by $\mathfrak{M}_1$ and $\mathfrak{M}_2$, since it vanishes for the integrals y_1 and y_2 , and hence is divisible by $[\mathfrak{M}_1, \mathfrak{M}_2]$; and since the order of the basis polynomial of $[\mathfrak{M}_1, \mathfrak{M}_2]$ must be at least 2, one has $\mathfrak{M} = [\mathfrak{M}_1, \mathfrak{M}_2]$. Further, $\mathfrak{M}_1$ and $\mathfrak{M}_2$ are relatively prime, because

$$\frac{y_2}{D}(y_1\xi - y_1') - \frac{y_1}{D}(y_2\xi - y_2') = 1, \quad D = D(y) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}. \quad (42)$$

They are also prime modules, since their residue group has order 1.

But besides y_1 and y_2 , $\mathfrak{M}$ also has all integrals

$$z_1 = \alpha_{11}y_1 + \alpha_{12}y_2, \quad z_2 = \alpha_{21}y_1 + \alpha_{22}y_2, \quad A = |\alpha_{ik}| \neq 0,$$

and hence is identical with the least common multiple of the two relatively prime modules $\mathfrak{N}_1$ and $\mathfrak{N}_2$, which have respectively the integrals z_1 and z_2 , with basis polynomials

$$N_i = z_i\xi - z_i' = (\alpha_{i1}y_1 + \alpha_{i2}y_2)\xi - (\alpha_{i1}y_1' + \alpha_{i2}y_2').$$

Thus $\mathfrak{M}$ admits infinitely many distinct representations as the least common multiple of relatively prime modules; by Theorem VII the two modules $\mathfrak{N}_1$ and $\mathfrak{N}_2$ are therefore of the same kind as one another and as all $\mathfrak{M}_1$ and $\mathfrak{M}_2$. Indeed, the residue groups of all these modules have order 1, and so are isomorphic.

We shall now write the additive decomposition of the residue group $\mathfrak{G} = \mathfrak{A}_1 + \mathfrak{A}_2$ corresponding to the representation (41). To do this we start, as in § 4, from relation (42), which expresses the relative primeness of $\mathfrak{M}_1$ and $\mathfrak{M}_2$. For the units a_1 and a_2 we obtain the residue classes generated by the polynomials

$$-\frac{y_1}{D}(y_2\xi - y_2') \quad \text{and} \quad \frac{y_2}{D}(y_1\xi - y_1').$$

Thus

$$\mathfrak{M}_1 = \left(\mathfrak{M}, \frac{y_2}{D}(y_1\xi - y_1') \right), \quad \mathfrak{M}_2 = \left(\mathfrak{M}, -\frac{y_1}{D}(y_2\xi - y_2') \right).$$

If we now form in the same way the units b_1 and b_2 corresponding to the decomposition $\mathfrak{M} = [\mathfrak{N}_1, \mathfrak{N}_2]$, replacing y_i by z_i , we get

$$\begin{aligned} -\frac{z_1}{D(z)}(z_2\xi - z_2') &= -\frac{\alpha_{11}y_1 + \alpha_{12}y_2}{AD(y)}((\alpha_{21}y_1 + \alpha_{22}y_2)\xi - (\alpha_{21}y_1' + \alpha_{22}y_2')), \\ \frac{z_2}{D(z)}(z_1\xi - z_1') &= \frac{\alpha_{21}y_1 + \alpha_{22}y_2}{AD(y)}((\alpha_{11}y_1 + \alpha_{12}y_2)\xi - (\alpha_{11}y_1' + \alpha_{12}y_2')), \end{aligned}$$

and therefore

$$\begin{cases} b_1 = \frac{\alpha_{11}y_1 + \alpha_{12}y_2}{y_1} \left(\frac{\alpha_{22}}{A} \right) a_1 + \frac{\alpha_{11}y_1 + \alpha_{12}y_2}{y_2} \left(-\frac{\alpha_{21}}{A} \right) a_2 = \frac{z_1}{y_1} \alpha^{11} a_1 + \frac{z_1}{y_2} \alpha^{12} a_2, \\ b_2 = \frac{\alpha_{21}y_1 + \alpha_{22}y_2}{y_1} \left(-\frac{\alpha_{12}}{A} \right) a_1 + \frac{\alpha_{21}y_1 + \alpha_{22}y_2}{y_2} \left(\frac{\alpha_{11}}{A} \right) a_2 = \frac{z_2}{y_1} \alpha^{21} a_1 + \frac{z_2}{y_2} \alpha^{22} a_2. \end{cases} \quad (43)$$

Here (α^{ik}) is the inverse matrix of (α_{ik}) .

Conversely, exchanging z_i with y_i gives

$$\begin{cases} a_1 = \frac{y_1}{z_1} \alpha_{11} b_1 + \frac{y_1}{z_2} \alpha_{12} b_2, \\ a_2 = \frac{y_2}{z_1} \alpha_{21} b_1 + \frac{y_2}{z_2} \alpha_{22} b_2. \end{cases} \quad (44)$$

These formulas display the isomorphism of the groups $\mathfrak{A}$ and $\mathfrak{B}$. Namely, by our general considerations, if $a_1 b_\sigma \neq 0$ the corresponding assignment is $a_1 \sim a_1 b_\sigma$, and hence by (44)

$$a_1 \sim \frac{y_1}{z_\sigma} \alpha_{1\sigma} b_\sigma, \quad \text{provided } \alpha_{1\sigma} \neq 0,$$

and by (43) likewise

$$b_\sigma \sim \frac{z_\sigma}{y_2} \alpha^{\sigma 2} a_2, \quad \text{provided } \alpha^{\sigma 2} \neq 0,$$

so that

$$\frac{y_1}{z_\sigma} \alpha_{1\sigma} b_\sigma \sim \frac{y_1}{z_\sigma} \alpha_{1\sigma} \frac{z_\sigma}{y_2} \alpha^{\sigma^2} a_2 = \frac{y_1}{y_2} \alpha_{1\sigma} \alpha^{\sigma^2} a_2.$$

Hence

$$a_1 \sim \alpha_{1\sigma} \alpha^{\sigma^2} \frac{y_1}{y_2} a_2 \quad \text{for } \alpha_{1\sigma} \alpha^{\sigma^2} \neq 0, \quad \frac{1}{\alpha_{1\sigma} \alpha^{\sigma^2}} \frac{y_2}{y_1} a_1 \sim a_2,$$

which gives the inverse. If (α_{ik}) is not a diagonal matrix, this condition $\alpha_{1\sigma} \alpha^{\sigma^2} \neq 0$ is always fulfilled for some σ . The case of a diagonal matrix must be excluded, since then the modules are identical. If, for example, only $\alpha_{12} = 0$, then $\mathfrak{A}_1 = \mathfrak{B}_1$, but not $\mathfrak{A}_2 = \mathfrak{B}_2$. Thus from $\mathfrak{A}_1 + \mathfrak{A}_2 = \mathfrak{B}_1 + \mathfrak{B}_2$ and $\mathfrak{A}_1 = \mathfrak{B}_1$ it does not follow that $\mathfrak{A}_2 = \mathfrak{B}_2$ (cf. note 12). The three modules $\mathfrak{M}_1$, $\mathfrak{M}_2$, and $\mathfrak{N}_2$ are then pairwise relatively prime, but do not form a totally relatively prime system, since $[\mathfrak{M}_1, \mathfrak{M}_2]$ is divisible by $\mathfrak{N}_2$ (cf. note 15).

Putting further

$$t_{12} = \alpha_{1\sigma} \alpha^{\sigma^2} \frac{y_1}{y_2} a_2, \quad t_{21} = \frac{1}{\alpha_{1\sigma} \alpha^{\sigma^2}} \frac{y_2}{y_1} a_1,$$

one verifies by direct calculation that the relations (35) are fulfilled.

2. Every module in more than one variable whose basis consists of a single polynomial gives an example of an infinite group. The following somewhat more complicated example is meant to show that infinite groups can also occur for multi-membered modules, that is, for modules whose basis contains more than one polynomial.²⁴

$$\mathfrak{M} = [(\xi + x\eta), (\xi + 1)] = [(P), (Q)] \quad \left(\xi = \frac{\partial}{\partial x}, \eta = \frac{\partial}{\partial y} \right).$$

The multiplication rule is that for differential expressions. The residue group is here infinite because the residue groups of $(\xi + x\eta)$ and $(\xi + 1)$ are both infinite, whereas the modules $(\xi + x\eta)$ and $(\xi + 1)$ are relatively prime, since

$$1 = (-x\xi - x^2\eta + 1)Q + (x\xi + x - 1)P.$$

It is now very easy to show that $\mathfrak{M}$ possesses no polynomial of the second degree: one sets

$$(u_1\xi + u_2\eta + u_3)(\xi + x\eta) = (v_1\xi + v_2\eta + v_3)(\xi + 1),$$

which gives $u_1 = u_2 = u_3 = v_1 = v_2 = v_3 = 0$. If one next looks for polynomials of the third degree in $\mathfrak{M}$, one finds two linearly independent ones, for example

$$(\xi^2 + x\xi\eta + \xi + (2 + x)\eta)Q = (\xi^2 + 2\xi + 1)P$$

and

$$(\xi^2 + 2x\xi\eta + x^2\eta^2 + \eta)Q = (\xi^2 + x\xi\eta + \xi + (x - 1)\eta)P.$$

If $\mathfrak{M}$ were a one-membered module, then both would have to be divisible by the basis polynomial, and, since $\mathfrak{M}$ contains no polynomial of the second degree, by a polynomial of the third degree; they would therefore have to be proportional, which they are not. The multi-memberedness of $\mathfrak{M}$ is the internal reason for the failure of Landau's product theorems, which are valid in the case of one variable.

3. A further example shows that infinite groups can also occur for prime modules.²⁵ Let P be the domain of all rational functions of two variables x, y . The module $\mathfrak{M}$ with basis

$$\xi - \frac{y}{x} = \xi - a$$

has an infinite group, since this group contains the residue class of every power of y ; on the other hand, $\mathfrak{M}$ is a prime module. Indeed, we show that every divisor

$$\mathfrak{M}_1 = (\xi - a, F(\xi, \eta))$$

is equal to the unit module. Every polynomial $F(\xi, \eta)$ can be represented modulo $(\xi - a)$ by a polynomial $G(\eta)$:

$$F(\xi, \eta) = c_0\eta^\nu + c_1\eta^{\nu-1} + \cdots + c_\nu \quad (\mathfrak{M}).$$

²⁴Cf. the example of Landau communicated by Blumberg (loc. cit., pp. 51–52). Here and below we denote the independent variables by x and y .

²⁵Our definition of “prime module” differs from what is called a prime module in the commutative case, since there divisors of lower dimension always exist, except when the dimension is 0, that is, when the residue group is finite.

Without loss of generality let $c_0 = 1$. By the multiplication law, $\xi\eta^\nu - \eta^\nu\xi = 0$; hence, since

$$\xi \equiv a(\mathfrak{M}_1), \quad \eta^\nu \equiv F_1(\mathfrak{M}_1),$$

where $F_1 = -(c_1\eta^{\nu-1} + \dots + c_\nu)$, one has

$$\xi F_1 - \eta^\nu a \equiv 0(\mathfrak{M}_1).$$

Expanding

$$\eta^\nu a = a\eta^\nu + \nu \frac{\partial a}{\partial y} \eta^{\nu-1} + \dots,$$

and again replacing η^ν by F_1 , one obtains

$$\xi(c_1\eta^{\nu-1} + \dots + c_\nu) - a(c_1\eta^{\nu-1} + \dots + c_\nu) + \nu \frac{\partial a}{\partial y} \eta^{\nu-1} + \dots \equiv 0(\mathfrak{M}_1),$$

or

$$c_1\eta^{\nu-1}a + \frac{\partial c_1}{\partial x}\eta^{\nu-1} + \dots - ac_1\eta^{\nu-1} + \dots + \nu \frac{\partial a}{\partial y} \eta^{\nu-1} + \dots \equiv 0(\mathfrak{M}_1),$$

or finally

$$\left(\frac{\partial c_1}{\partial x} + \nu \frac{\partial a}{\partial y} \right) \eta^{\nu-1} + \dots \equiv 0(\mathfrak{M}_1).$$

On the left now stands a polynomial in η of exact degree $\nu - 1$ which does not vanish identically, for the highest coefficient

$$\frac{\partial c_1}{\partial x} + \nu \frac{\partial a}{\partial y} = \frac{\partial c_1}{\partial x} + \nu \frac{1}{x} \quad \text{is} \neq 0,$$

since otherwise $c_1 = -\nu \log x + \varphi(y)$ would not be rational. Thus the procedure can be continued and leads to a non-identically-vanishing polynomial of degree zero, proving that the unit is also contained in $\mathfrak{M}_1$.²⁶

4. On the other hand, if one adjoins the function $\log x$ to the rationality domain, then $\mathfrak{M}$ has the divisors

$$\left(\xi - \frac{y}{x}, \eta - \log x + \varphi(y) \right),$$

where $\varphi(y)$ runs through all rational functions of y . These divisors are all distinct, the integrability condition is fulfilled for each, and the corresponding integral is

$$y \log x - \int \varphi(y) dy + c.$$

Therefore the unit cannot be contained in the module, since otherwise the zero function would be the only possible integral. On the other hand, ξ and η are congruent, with respect to the module, to a quantity of the domain; hence the residue group is finite and of order 1. Further, any two of the infinitely many divisors are relatively prime, and the given module $\mathfrak{M}$ is equal to the least common multiple $\mathfrak{N}$ of all these divisors. Indeed, it is divisible by all divisors, hence also by $\mathfrak{N}$. If $\mathfrak{N}$ were a proper divisor of $\mathfrak{M}$, then $\mathfrak{N}$ would still have to contain at least one polynomial F , say of degree ρ , depending only on η alone. Letting $\varphi(y)$ run through the functions $y, \dots, y^{\rho+1}$, there is no linear relation with coefficients independent of y between the corresponding integrals

$$y \log x - \frac{y^{i+1}}{i+1} + c_i \quad (i = 1, \dots, \rho + 1).$$

But the differential equation of order ρ , $F = 0$, cannot possess $\rho + 1$ linearly independent integrals.

The module $\mathfrak{M} = \mathfrak{N}$ is, however, not completely reducible.²⁷ For otherwise it would be the least common multiple of finitely many of these prime modules, and its residue group would therefore be of finite order; yet the residue group of $\xi - y/x$ contains the residue classes of all powers of η , and so is infinite.

Göttingen, August 1, 1919.

(Received August 4, 1919.)

²⁶This procedure amounts to proving that the integrability conditions necessary for a system of differential equations are not fulfilled.

²⁷Thus we have an example of the first case of Theorem X.

18. On a Work of K. Hentzelt, Fallen in the War, on Elimination Theory

Jahresbericht der DMV 30 (1921), p. 101

11th meeting, Friday, September 23, 1921, 11 a.m.

(Chair: Loewy.)

1. E. Noether: On a work of K. Hentzelt, fallen in the war, on elimination theory.

Kurt Hentzelt has been missing before Dixmuiden since October 1914 and must be counted among the dead. I intend soon to publish, in a free redaction in the *Mathematische Annalen*, the essential parts of his Erlangen dissertation, which he left behind built up without gaps but almost incomprehensibly presented.

The matter concerns the general problem of **elimination theory**: to determine the common zeros of all polynomials of an ideal $\mathfrak{M}$ consisting of polynomials. If the variables are first subjected to a linear transformation with indeterminate coefficients, then the main result can be stated as follows:

To the ideal $\mathfrak{M}$ there can be assigned uniquely a resultant form

$$R_{\mathfrak{M}} = R^{(1)}(x_1, \dots, x_n) \cdots R^{(i)}(x_i, \dots, x_n) \cdots R^{(m)}(x_n) = 0(\mathfrak{M}),$$

such that $R_{\mathfrak{M}}$ vanishes for all zeros of $\mathfrak{M}$ and only for these. If $\mathfrak{N}$ is a divisor of $\mathfrak{M}$ and $R_{\mathfrak{N}}$ and $R_{\mathfrak{M}}$ agree, then $\mathfrak{N}$ and $\mathfrak{M}$ themselves also agree.

The second part of the theorem shows that the resultant form supplies the zeros in characteristic multiplicity. It rests on the fact that $\mathfrak{M}$ can be regarded as a module of linear forms in the power products of $x_1, \dots, x_{i-1}$; the individual factors $R^{(i)}(x_i, \dots, x_n)$ of $R_{\mathfrak{M}}$ then become the norms of the respective “basic module” with respect to this module, when $x_{i+1}, \dots, x_n$ is adjoined to the coefficient domain. If one invokes abstract ideal theory, this yields the following interpretation: if

$$\mathfrak{M} = [\mathfrak{Q}, \mathfrak{Q}_1, \dots, \mathfrak{Q}_r] = [\mathfrak{Q}, \mathfrak{L}]$$

is a decomposition of $\mathfrak{M}$ into primary ideals $\mathfrak{Q}_i$, and $\mathfrak{L}$ is the complement of $\mathfrak{Q}$, then to the ideal $\mathfrak{Q}$ there corresponds a primary factor $Q^{(i)}(x_i, \dots, x_n)$ of $R^{(i)}(x_i, \dots, x_n)$, which, in the sense specified above, represents the norm of $\mathfrak{L}$ with respect to $\mathfrak{Q}$.